

End –End Communication at Transport Layer

- a) Implement echo client server using TCP/UDP sockets.
- b) Implement a chat program using socket programming.

AIM: To implement (a) an echo client-server using TCP and UDP sockets, and (b) a chat program using socket programming in Python.

tcp_server.py

```
import socket

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

server.bind(("127.0.0.1", 5000))

server.listen(1)

print("TCP Echo Server started...")

conn, addr = server.accept()

print("Connected:", addr)

while True:

    data = conn.recv(1024)

    if not data:

        break

    print("Client:", data.decode())

    conn.send(data)

conn.close()

server.close()
```

tcp_client.py

```
import socket

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect(("127.0.0.1", 5000))

print("Connected to TCP Echo Server")

while True:
    msg = input("Enter message: ")
    if msg.lower() == "exit":
        break

    client.send(msg.encode())
    data = client.recv(1024)

    print("Server:", data.decode())

client.close()
```

udp_server.py

```
import socket

server = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
server.bind(("127.0.0.1", 5001))

print("UDP Echo Server started...")

while True:
    data, addr = server.recvfrom(1024)
    print("Client:", data.decode())

    if data.decode().lower() == "exit":
        break

    server.sendto(data, addr)

server.close()
```

udp_client.py

```
import socket

client = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)

print("Connected to UDP Echo Server")

while True:
    msg = input("Enter message: ")

    client.sendto(msg.encode(), ("127.0.0.1", 5001))

    if msg.lower() == "exit":
        break

    data, addr = client.recvfrom(1024)
    print("Server:", data.decode())

client.close()
```

chat_server.py

```
import socket

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

server.bind(("127.0.0.1", 5002))

server.listen(1)

print("Chat Server started...")

conn, addr = server.accept()

print("Connected:", addr)

while True:

    msg = conn.recv(1024).decode()

    if msg.lower() == "exit":

        break

    print("Client:", msg)

    reply = input("You: ")

    conn.send(reply.encode())

    if reply.lower() == "exit":

        break

conn.close()

server.close()
```

chat_client.py

```
import socket
```

```
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
```

```
client.connect(("127.0.0.1", 5002))
```

```
print("Connected to Chat Server")
```

```
while True:
```

```
    msg = input("You: ")
```

```
    client.send(msg.encode())
```

```
    if msg.lower() == "exit":
```

```
        break
```

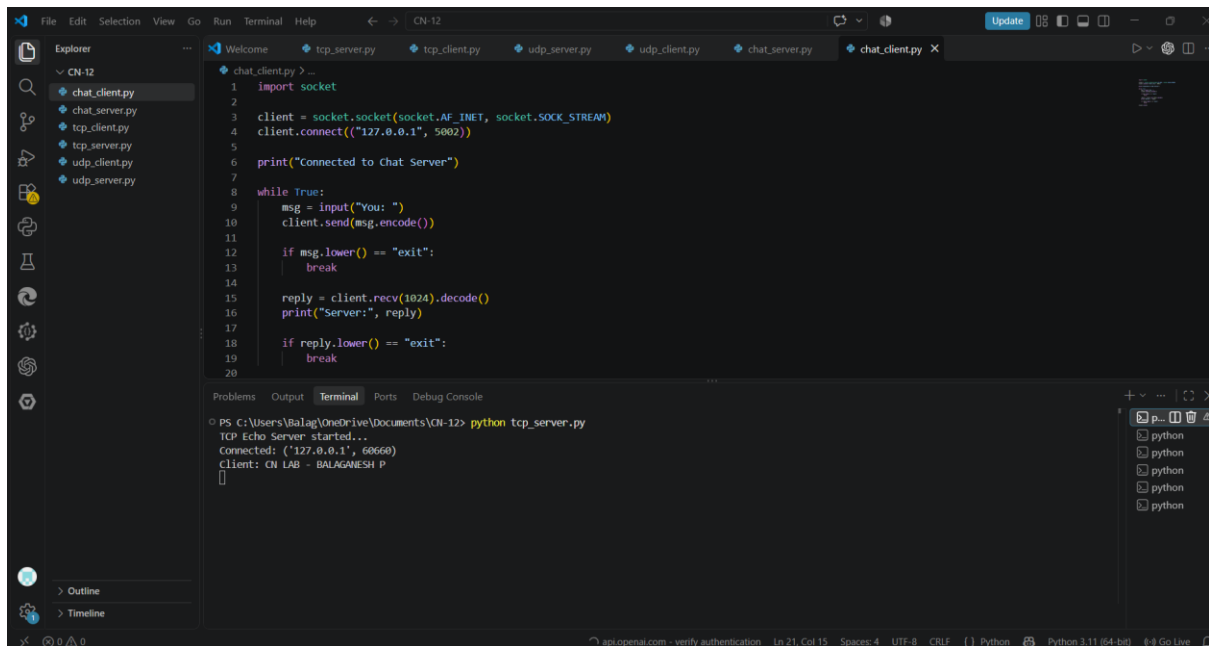
```
    reply = client.recv(1024).decode()
```

```
    print("Server:", reply)
```

```
    if reply.lower() == "exit":
```

```
        break
```

```
client.close()
```



The screenshot shows the Visual Studio Code editor interface. The Explorer sidebar on the left displays a file tree for a project named 'CN-12', containing files: chat_client.py, chat_server.py, tcp_client.py, tcp_server.py, udp_client.py, and udp_server.py. The main editor window has several tabs open, with 'chat_client.py' selected. The code in this file is as follows:

```
1 import socket
2
3 client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
4 client.connect(("127.0.0.1", 5002))
5
6 print("Connected to Chat Server")
7
8 while True:
9     msg = input("You: ")
10    client.send(msg.encode())
11
12    if msg.lower() == "exit":
13        break
14
15    reply = client.recv(1024).decode()
16    print("Server:", reply)
17
18    if reply.lower() == "exit":
19        break
20
```

Below the code editor, the 'Terminal' tab is active, showing the following output:

```
PS C:\Users\Balag\OneDrive\Documents\CN-12> python tcp_server.py
TCP Echo Server started...
Connected: ('127.0.0.1', 60660)
Client: CN LAB - BALAGANESH P

```

Problems Output **Terminal** Ports Debug Console

```
PS C:\Users\Balag\OneDrive\Documents\CN-12> python tcp_client.py
Connected to TCP Echo Server
Enter message: CN LAB - BALAGANESH P
Server: CN LAB - BALAGANESH P
Enter message: 
```

Problems Output **Terminal** Ports Debug Console

```
PS C:\Users\Balag\OneDrive\Documents\CN-12> python udp_server.py
UDP Echo Server started...
Client: CN LAB EXP

```

```
PS C:\Users\Balag\OneDrive\Documents\CN-12> python udp_client.py
Connected to UDP Echo Server
Enter message: CN LAB EXP
Server: CN LAB EXP
Enter message: 
```

```
PS C:\Users\Balag\OneDrive\Documents\CN-12> python chat_server.py
Chat Server started...
Connected: ('127.0.0.1', 58511)
Client: HII
You: GOOD MORNING
Client: WELCOME
You: 
```

Problems Output **Terminal** Ports Debug Console

```
PS C:\Users\Balag\OneDrive\Documents\CN-12> python chat_client.py
Connected to Chat Server
You: HII
Server: GOOD MORNING
You: WELCOME

```